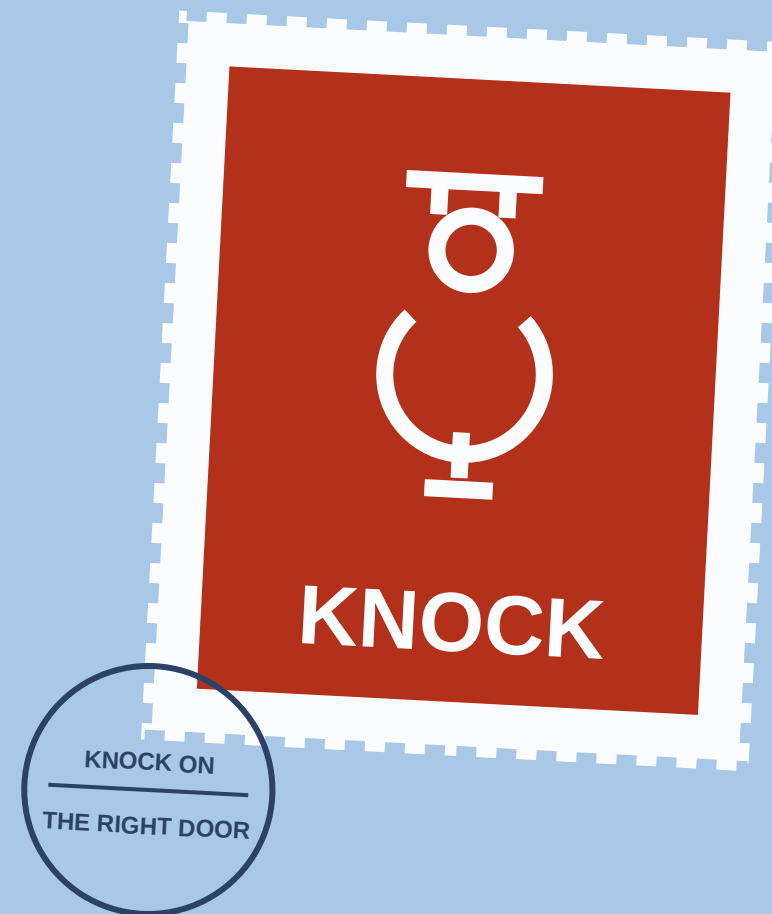


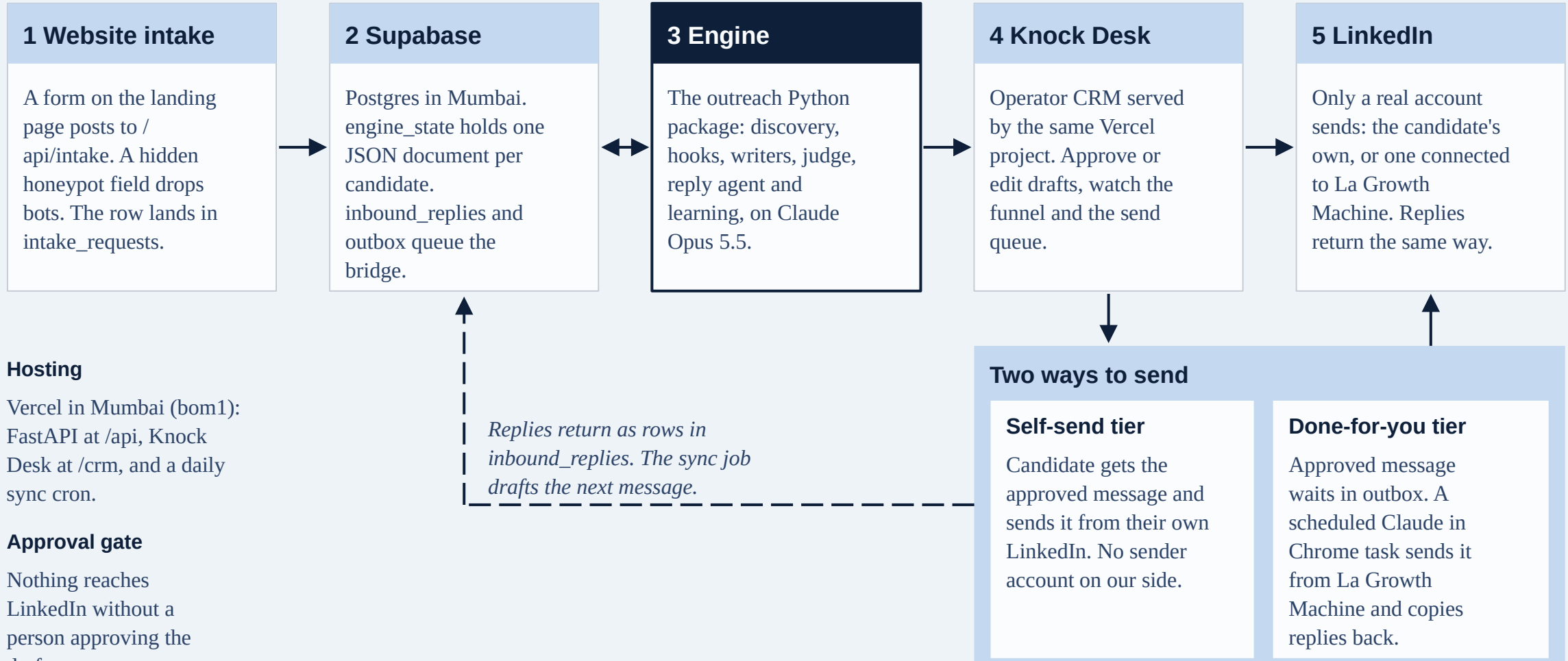
KNOCK BACKEND · PRODUCT AND TECHNOLOGY

No citation, no hook.

How the engine finds the right people at a candidate's target companies, researches each one, and writes the first message. A person approves it before it sends.



A sign-up flows through five parts, and nothing reaches LinkedIn without a person saying yes



Two layers of state: JSON documents the engine runs on, and a relational schema built to be queried

Runtime tables, used by the engine today

engine_state

One JSON document per candidate, plus shared feedback and usage documents.

inbound_replies

Replies copied from the LGM inbox. The engine stamps processed_at and result.

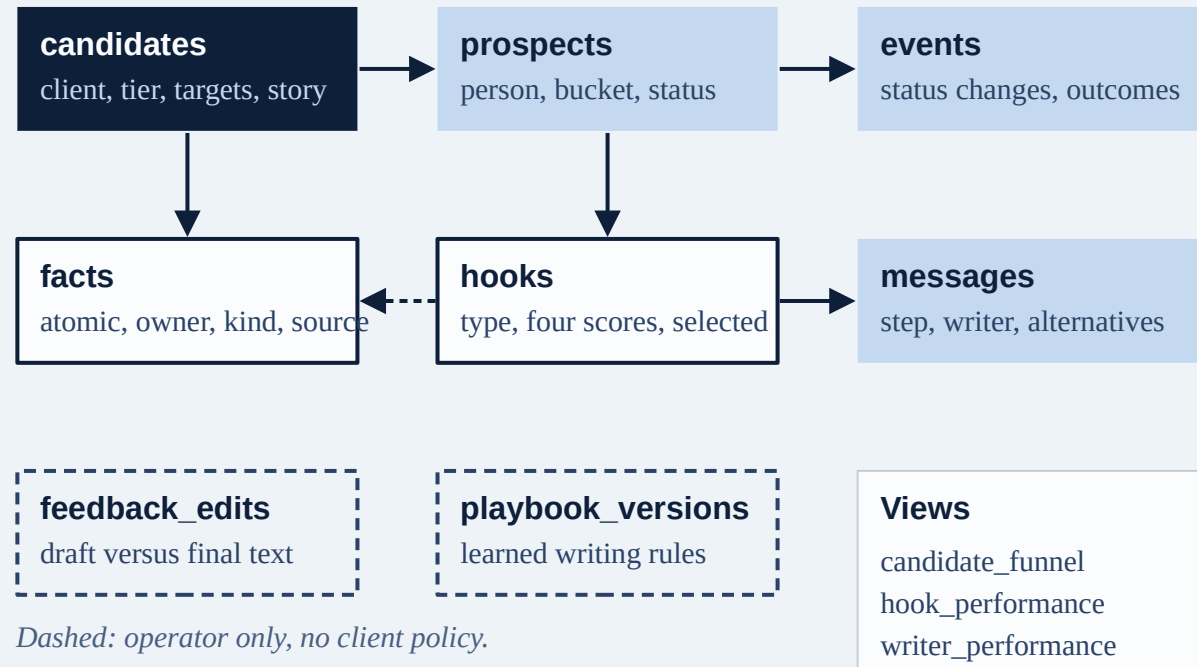
outbox

Messages waiting to send, with a send_after time. The sender stamps sent_at.

intake_requests

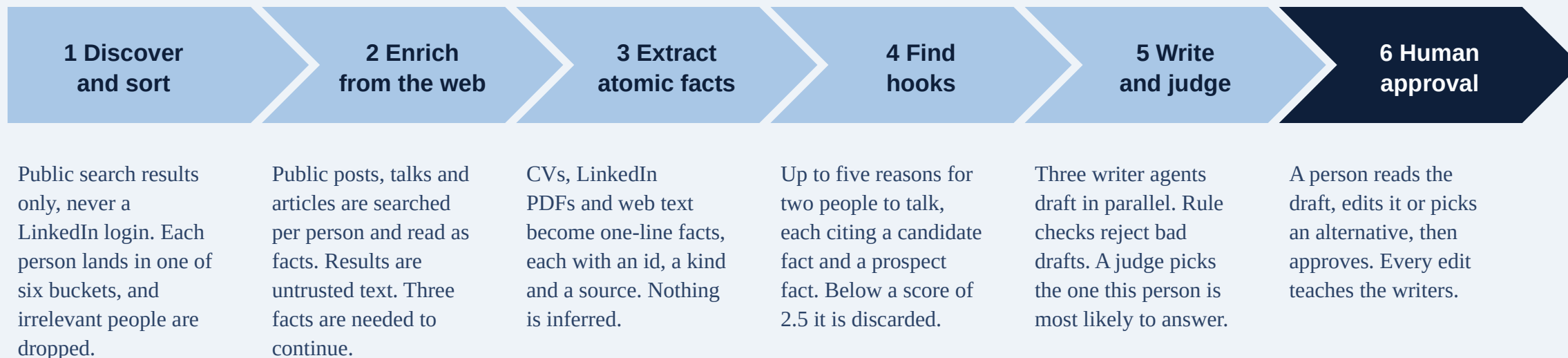
Website sign-ups. Anyone can insert; only the engine can read them.

Relational core (migration 0001), ready for analytics and client access

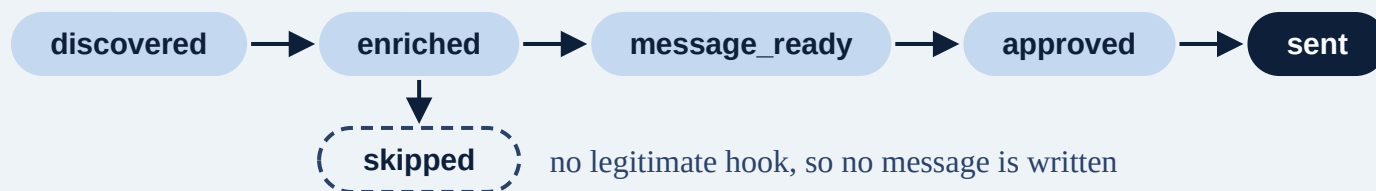


The schema enforces the design. hooks.candidate_fact_id and hooks.prospect_fact_id are both required references to facts(id), so the database refuses a hook that cites nothing. Row-level security is on for every table.

Six stages turn a company name into a reviewed message, and any stage can drop a prospect



PROSPECT STATUS AS IT MOVES



The system prefers fewer, better messages. Stopping early is designed in.

A hook exists only if two real facts prove it, and four weighted scores decide which one is used

How a hook is built (the facts here are illustrative)

CANDIDATE FACT c:3

Moved from consumer brands into product in 2024.

PROSPECT FACT p:rahul:1

Moved from Flipkart (category manager) to Zepto (product manager) in 2024.

HOOK · similar_transition

One line on why these two people have a real reason to talk.

candidate_fact_id c:3
prospect_fact_id p:rahul:1

No citation, no hook.

Four scores, rated 1 to 5, one weight each

Specificity

0.30

How concrete and checkable the overlap is.

Rarity

0.25

How few people share it. A small team beats a big city.

Relevance

0.30

How closely it relates to the target role at this company.

Recency

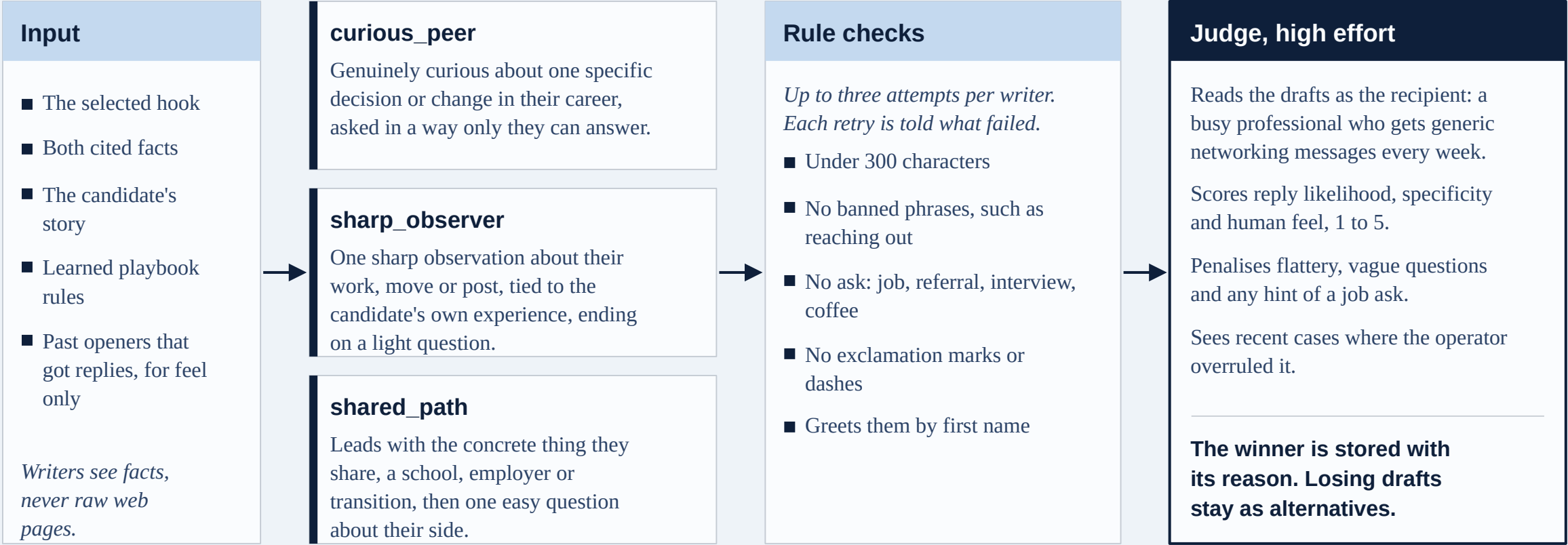
0.15

How recent the prospect's side is. The last six months scores highest.

$$\text{score} = 0.30 \times \text{specificity} + 0.25 \times \text{rarity} + 0.30 \times \text{relevance} + 0.15 \times \text{recency}$$

- Minimum score is 2.5 of 5. Below it the hook is not used.
- The top hook is selected; a second one powers the follow-up.
- Seven types: shared school, shared employer, similar transition, relevant work, post reaction, shared geography, shared interest.

Three writers compete, rule checks filter, and a judge picks the draft this person is most likely to answer



Fallbacks: if no draft passes the rules, the first one returns flagged for manual review. If only one passes, it wins without a judge.

Replies are read as a whole thread, and an ask ladder decides when it is right to ask for more

From reply to draft

1 Reply copied in

Claude in Chrome adds it to inbound_replies.



2 Matched to a prospect

By profile URL, else a unique name. Duplicates skipped.



3 Read as a thread

Sentiment, rapport 1 to 5, redirect, and the right ask.



4 Next message drafted

Under 400 characters, answers them first, one ask at most.



5 Auto-send gate

Passes the safety filter, or is held for a person.

The ask ladder, ordered by how much it asks (our reading)

none

Rapport below 3. Another real exchange first.

advice

Ask how best to approach applying.

opening

Ask if a relevant role exists on their team.

hiring_manager

Ask who owns hiring for the role.

intro

Ask for an introduction to someone they named.

referral

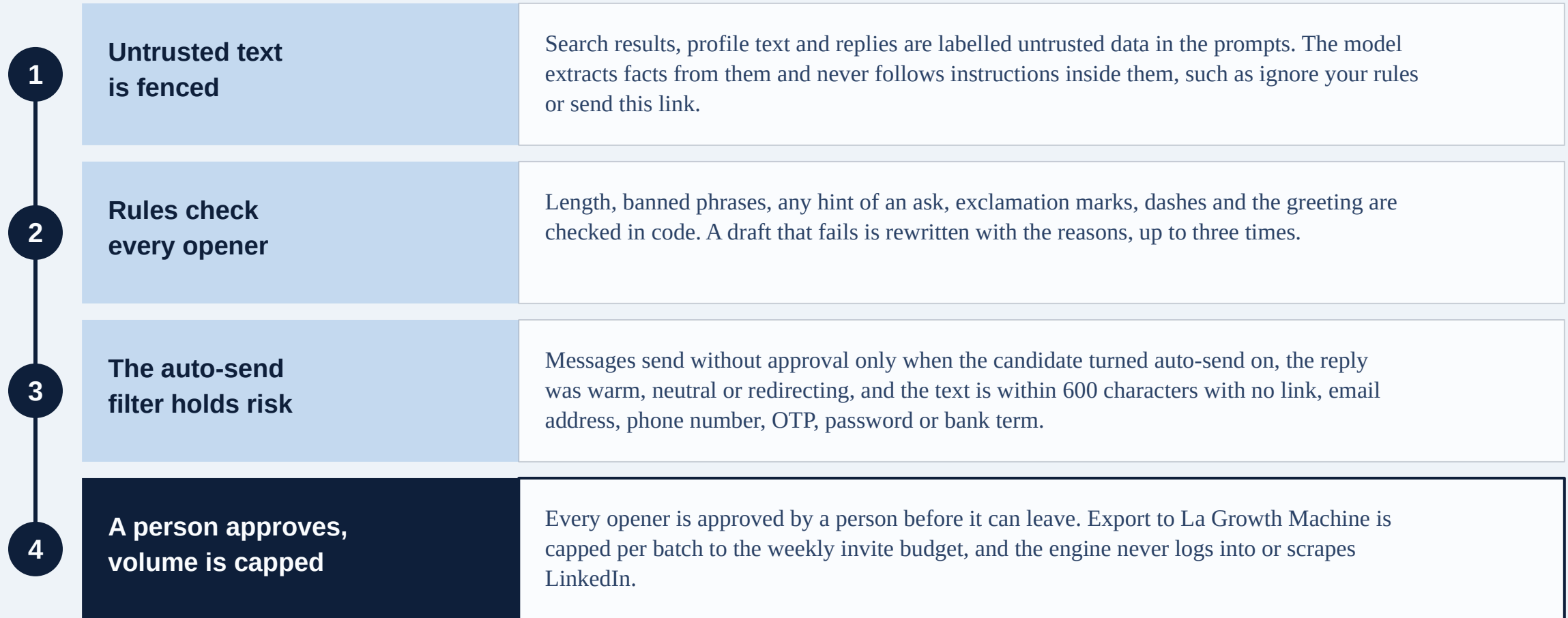
Only at rapport 4+ with a specific role.

Follow-ups: after six silent days, one short note built on a second verified fact, under 220 characters.
If both messages go unanswered, the thread closes as no_response. Reply text is untrusted data.

Every edit and every outcome feeds back, so the writers, the judge and the reply agent improve



Four layers stand between model output and a LinkedIn inbox, and the last one is a person



Small on purpose: one FastAPI function, one Postgres project and one scheduled job, all in Mumbai

Vercel, Mumbai (bom1)

POST /api/intake

Public sign-up form, validated, with a honeypot field

/api/crm/*

Knock Desk API, behind the operator password

GET /api/cron/sync

Daily at 09:00 IST, behind a bearer secret

/crm

Knock Desk, a single page with seven tabs

One function, 60 second limit, engine copied in at build.

REST
token

Supabase, Mumbai (ap-south-1)

- Postgres with row-level security on every table.
- The engine uses the publishable key plus a secret x-engine-token header, so no service-role key is kept.
- Only the SHA-256 hash of the token is stored, and engine_authorized() checks it inside each policy.
- The public can insert sign-ups and nothing else.
- Same city as the function, so calls stay short.

Settings, by name only

SUPABASE_URL

Project address

SUPABASE_KEY

Publishable key only

ENGINE_TOKEN

Gates row access

CRM_PASSWORD

Knock Desk login

CRON_SECRET

Authorises the cron

ANTHROPIC_API_KEY

Needed for sync only

Knock Desk tabs

Overview

Candidates

Send queue

Replies

Sign-ups

Learning

Spend

Tabs show a count when drafts or sign-ups wait for you.

The design is tested twice: a blind comparison of writing approaches, and a suite that needs no model

Blind message test: seven approaches, the same test pairs

Knock engine: 3 writers and a judge	Hybrid playbook with self-judge
Research playbook	Single-writer engine
Sales copywriting	Five-point networking method
Plain baseline	

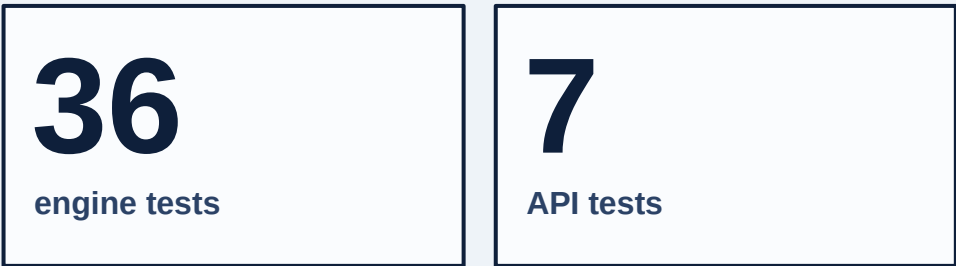
Ten test pairs. Four blind judges: a recipient, a hiring manager, an Indian recruiter and a rubric auditor.

Our engine beat the six other approaches.

What the test changed in prompts.py

- Openers: no self-introduction, concrete batch-year ties, and a practitioner question only that person could answer.
- Follow-ups: a second verified fact, an insider line, either/or questions, an openly smaller ask, an easy out.
- Three writers plus a judge stays the default. It beat the single-writer version of the same engine.

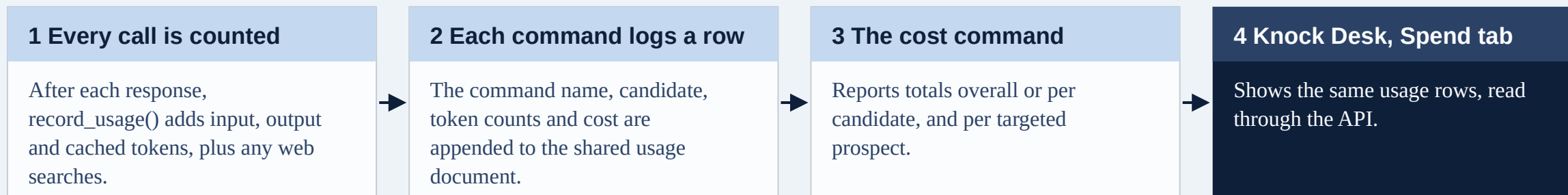
Automated tests, no model needed



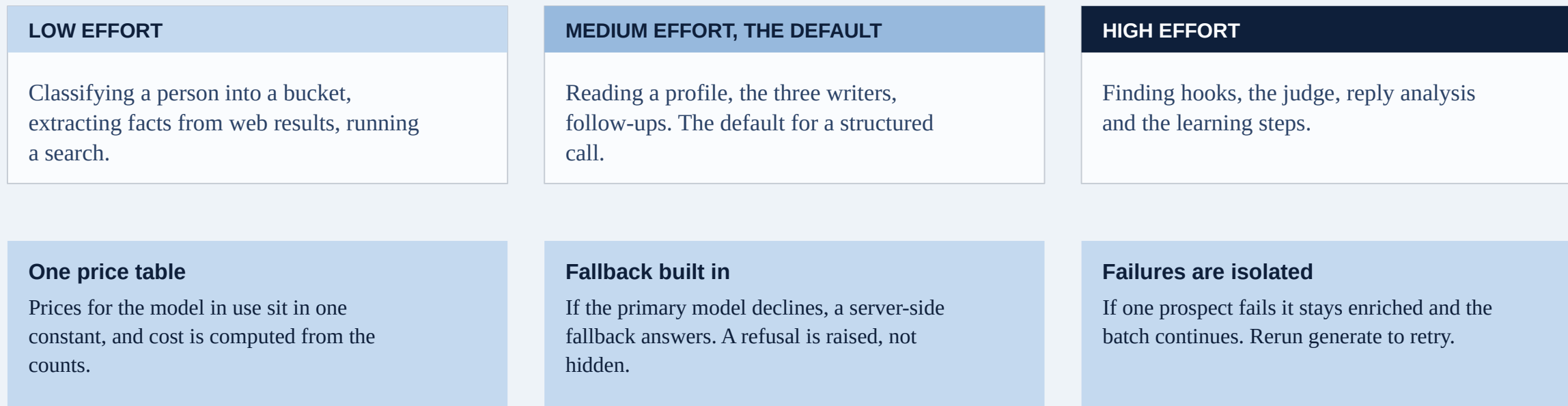
Both pass on the last run, offline, no model calls.

- No citation, no hook: invented fact ids are dropped.
- Opener rules reject asks, spam phrases, dashes and a missing name.
- A reply that tries to inject a link is never auto-sent.
- Thompson sampling favours winners and still tries new writers.
- Weak writers retire and a new one is spawned.
- Follow-ups draft once, then the thread closes.
- The API demands the password and the cron secret.

Every model call is metered, so spend is measured per command, candidate and prospect, never guessed



Effort is matched to the task



The engine and desk are built and tested; three open steps stand between today and a live end-to-end run

Built and tested

- Engine: discovery, hooks, writers, judge, reply agent and learning.
- 36 engine tests and 7 API tests pass.
- Supabase schema applied, with row-level security and the token gate.
- FastAPI and the Knock Desk are set up on Vercel.
- Website sign-ups land in intake_requests and import as candidates.
- Cost metering and the cost command are in place.

Open before go-live (dashed means not yet done)

- 1 Add ANTHROPIC_API_KEY in Vercel**
Until it is set, sync and the daily cron answer with an error, and no replies are drafted.
- 2 Run one real candidate end to end**
Every hook and message is reviewed by hand and every edit is logged. The first real learning signal.
- 3 Schedule the LGM bridge task**
The Claude in Chrome task that sends due outbox rows and copies replies back is written, not yet scheduled.

The open steps are a setting, a run and a schedule.

Next: close the loop on a real candidate, then widen the sender and the client view

Now

- Set ANTHROPIC_API_KEY in Vercel.
- Schedule the LGM bridge task.
- Run one real candidate end to end, reviewing every hook and message.

Next

- Tune the prompts from the edits that run produces.
- Let the first learning cycle run once five signals exist.
- Set safe sending limits per account.
- Read funnel views by bucket, hook type and company.

Later

- Swap Chrome for the LGM API. lgm.py is ready; it needs the Pro plan.
- Receive accepted and replied events through a webhook.
- Open a client site with magic-link login, on the row-level security already written.

THREE DESIGN PRINCIPLES TO HOLD ON TO

Cited

No citation, no hook.

Approved

A person says yes before anything sends.

Learning

Edits and outcomes rewrite the rules.